Sahaj B Bhadja

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Summary

Software Development Engineer with 2+ years of experience building **scalable Java/Python services**, **distributed systems**, cloud applications, automated tests, and reliable customer-focused software across production environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Achieved 99%+ reliability across 5K+ monthly jobs through **validation, reconciliation, and error handling**, cloud storage, CI/CD releases, and production monitoring supporting dependable distributed workflows reliably at scale.
- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant microservices** with explicit state transitions and failure isolation, preserving recoverable background processing under sustained production workloads reliably.
- Reduced dashboard latency from 1.2s to 350ms by optimizing **indexed queries and normalized PostgreSQL schemas**, supporting 500+ users with responsive analytics and row-level tenant isolation reliably at scale.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, secure ingestion, AWS RDS, and CI/CD workflows.
- Processed 1K+ files per batch using **multithreaded services and fault isolation**, validating heterogeneous inputs while preserving responsive customer workflows during concurrent model execution and sustained production traffic reliably.
- Maintained 90%+ scoring consistency across 10K+ submissions by implementing **structured model evaluation**, fallback paths, persistent results, and reviewer controls for reproducible grading and reliable production analytical workflows.

USC AD Labs | *ML Research Assistant* | *Los Angeles, CA* **January 2025 – May 2025**
- Improved long-horizon trajectory planning by 18% in simulation by building a **reproducible evaluation pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and controlled experiments across diverse scenarios reliably.
- Improved closed-loop planning stability by 15% through a **simulation-based validation model** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across complex scenarios reliably.
- Built **transformer-ready data pipelines** encoding five-second ego trajectories at 10Hz, HD-map vectors, nearby-agent context, and aligned coordinate features for controlled offline reinforcement-learning experiments and reproducible closed-loop evaluation.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments under sustained traffic.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.
- Increased ETL efficiency by 68% using **AWS Glue and S3** to transform JSON, CSV, and Parquet inputs into normalized datasets, reducing storage costs by 15% reliably at scale.

Skills

Languages: Java, Python, C++, TypeScript, JavaScript, SQL, Bash
Computer Science: Data Structures, Algorithms, Object-Oriented Programming, Design Patterns, Distributed Systems, Concurrent Programming, Multithreading, System Design
Backend & Data: REST APIs, Microservices, Event-Driven Architecture, JSON, Caching, PostgreSQL, MySQL, MongoDB, AWS RDS, NoSQL, Data Modeling, Query Optimization
Cloud & Delivery: AWS, Azure, GCP, Docker, Kubernetes, Jenkins, Git, GitHub, CI/CD, Maven, Linux, Build Automation
Quality & Operations: Unit Testing, Integration Testing, Automated Testing, Regression Testing, PyTest, Mockito, Debugging, Troubleshooting, Monitoring, Observability, Error Handling, Fault Isolation, Performance Optimization, Production Reliability
Engineering Practices: Agile, Code Review, Requirements Analysis, Technical Documentation, Software Development Lifecycle, Root-Cause Analysis, Cross-Functional Collaboration, AI-Assisted Development

Projects

Distributed Order Processing Service - Design Concept
- Designed a **Java/AWS distributed-service concept** using queues, idempotent consumers, DynamoDB, caching, and autoscaling to process customer orders reliably, recover from partial failures, and expose detailed operational metrics.

GenAI Code Validation Pipeline - Design Concept
- Designed a **Python AI-assisted development pipeline concept** that generates candidate changes, runs static analysis and automated tests, records review evidence, detects unsafe output, and requires human approval before integration.

Event-Driven Customer Notification Platform - Design Concept
- Designed a **TypeScript event-driven platform concept** using REST APIs, asynchronous workers, retry policies, dead-letter queues, and observability dashboards to deliver customer notifications reliably across high-volume production workloads.